

KON 313

Feedback Control Systems

Modeling of the Control Systems in Frequency Domain

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Developing a Mathematical Model

- Deriving reasonable mathematical models is the most important part of the entire analysis of control systems.
- A mathematical model of a dynamic system is defined as a set of equations that represents the dynamics of the system accurately, or at least fairly well.
- The dynamics of many systems, whether they are mechanical, electrical, biological, and so on, may be described in terms of differential equations.
- Such differential equations may be obtained by using physical laws governing a particular system—for example, Newton's laws for mechanical systems and Kirchhoff's laws for electrical systems.

Kirchhoff's voltage law

The sum of voltages around a closed path equals zero.

Kirchhoff's current law

The sum of electric currents flowing from a node equals zero.

Newton's laws

The sum of forces on a body equals zero;³ the sum of moments on a body equals zero.

Developing a Mathematical Model

- Applying the laws of physics lead to mathematical models that describe the relationship between the input and output of dynamic systems.
- One such model is the linear, time-invariant differential equation:

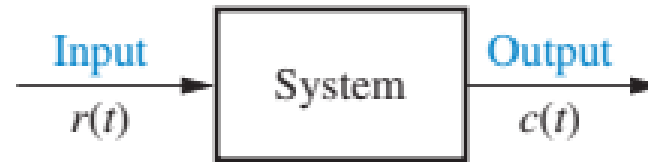
$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \cdots + a_0 c(t) = \\ b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \cdots + b_0 r(t)$$

which relates the output, $c(t)$, to the input, $r(t)$, by way of the system parameters, a_i and b_j .

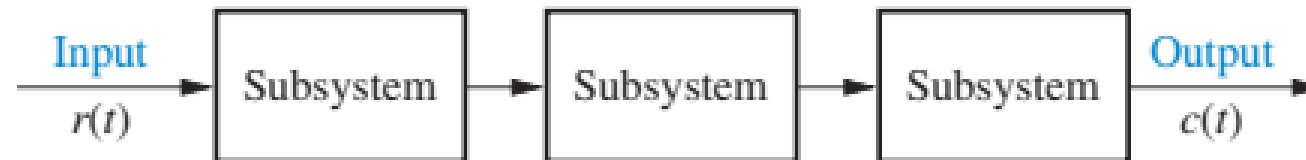
The differential equation is not a satisfying representation from a system perspective, since the system parameters, the output, and the input appear throughout the equation.

Developing a Mathematical Model

- We would prefer a mathematical representation where the input, output, and system are distinct and separate parts.



- Also, we would like to represent conveniently the interconnection of several subsystems. For example, cascaded interconnections:



- where a mathematical function, called a transfer function, is inside each block, and block functions can easily be combined.

Laplace Transform Review

- With Laplace transform, we can represent the input, output, and system as separate entities.
- The Laplace transform is defined as

$$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$$

where $s = \sigma + j\omega$ is a complex value.

- The inverse Laplace transform, which allows us to find $f(t)$ given $F(s)$

$$\mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds = f(t)u(t)$$

where $u(t)$ is the unit step function. Multiplication of $f(t)$ by $u(t)$ yields a time function that is zero for $t < 0$.

$$\begin{aligned} u(t) &= 1 & t > 0 \\ &= 0 & t < 0 \end{aligned}$$

Laplace Transform Review

- Laplace Transform Table

Item no.	$f(t)$	$F(s)$	Description
1.	$\delta(t)$	1	Impulse (Dirac-delta)
2.	$u(t)$	$\frac{1}{s}$	Step
3.	$tu(t)$	$\frac{1}{s^2}$	Ramp
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$	Parabolic
5.	$e^{-at}u(t)$	$\frac{1}{s+a}$	Exponential
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$	Sinusoidal
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$	

Laplace Transform Review

- Laplace Transform Theorems

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s + a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t - T)] = e^{-sT} F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a} F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^n s^{n-k} f^{(k-1)}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_{0-}^t f(\tau) d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²

Laplace Transform Review

Example – 1 (Laplace transform)

- Find the Laplace transform of $f(t) = Ae^{-at}u(t)$
- Since the time function does not contain an impulse function, we can replace the lower limit of the integral in the Laplace transform definition with 0.
Hence,

$$\begin{aligned} F(s) &= \int_0^{\infty} f(t)e^{-st} dt = \int_0^{\infty} Ae^{-at}e^{-st} dt = A \int_0^{\infty} e^{-(s+a)t} dt \\ &= -\frac{A}{s+a} e^{-(s+a)t} \Big|_{t=0}^{\infty} = \frac{A}{s+a} \end{aligned}$$

Example – 2 (Inverse Laplace transform)

- Find the inverse Laplace transform of

$$F_1(s) = 1/(s + 3)^2$$

- Solution:**

- For this example we make use of the **frequency shift theorem** and the Laplace transform of $f(t) = tu(t)$
- The inverse transform of $F(s) = 1/s^2$ is $f(t) = tu(t)$ and
- The inverse transform of $F(s + a) = 1/(s + a)^2$ is $e^{-at}tu(t)$.
- Hence,

$$f_1(t) = e^{-3t}tu(t)$$

Partial Fraction Expansion

- To find the inverse Laplace transform of a complicated function, we can convert the function to a sum of simpler terms for which we know the Laplace transform of each term. The result is called a partial-fraction expansion.
- If $F(s) = N(s)/D(s)$ is **proper** where the order of $N(s)$ is less than the order of $D(s)$, then a partial-fraction expansion can be made.
- If $F(s)$ is **improper** where the order of $N(s)$ is greater than or equal to the order of $D(s)$, then $N(s)$ must be divided by $D(s)$ successively until the result has a remainder whose numerator is of order less than its denominator. For example,

$$F_1(s) = \frac{s^3 + 2s^2 + 6s + 7}{s^2 + s + 5}$$

$$F_1(s) = s + 1 + \frac{2}{s^2 + s + 5}$$

$$f_1(t) = \frac{d\delta(t)}{dt} + \delta(t) + \mathcal{L}^{-1}\left[\frac{2}{s^2 + s + 5}\right]$$

Partial Fraction Expansion

- Three cases might happen in which the roots of $D(s)$ are:
 1. Real and Distinct
 2. Real and Repeated
 3. Complex/Imaginary

Partial Fraction Expansion

Case 1. Roots of the Denominator of $F(s)$ Are Real and Distinct

Partial fraction expansion of $F(s)$ is written as a sum of terms where each factor of the original denominator forms the denominator of each term, and constants, called residues, form the numerators.

$$\begin{aligned} F(s) &= \frac{N(s)}{D(s)} \\ &= \frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_m)\cdots(s+p_n)} \\ &= \frac{K_1}{(s+p_1)} + \frac{K_2}{(s+p_2)} + \cdots + \frac{K_m}{(s+p_m)} + \cdots + \frac{K_n}{(s+p_n)} \end{aligned}$$

To evaluate each residue, K_i , we multiply the equation by the denominator of the corresponding partial fraction.

$$\begin{aligned} (s+p_m)F(s) &= \frac{(s+p_m)N(s)}{(s+p_1)(s+p_2)\cdots(s+p_m)\cdots(s+p_n)} \\ &= (s+p_m)\frac{K_1}{(s+p_1)} + (s+p_m)\frac{K_2}{(s+p_2)} + \cdots + K_m + \cdots \\ &\quad + (s+p_m)\frac{K_n}{(s+p_n)} \end{aligned}$$

If we let s approach $-P_m$:

$$\frac{\cancel{(s+p_m)}N(s)}{(s+p_1)(s+p_2)\cdots\cancel{(s+p_m)}\cdots(s+p_n)} \Big|_{s \rightarrow -p_m} = K_m$$

Partial Fraction Expansion

Case 1. Roots of the Denominator of $F(s)$ Are Real and Distinct

Ex.

$$F(s) = \frac{2}{(s+1)(s+2)}$$

$$F(s) = \frac{2}{(s+1)(s+2)} = \frac{K_1}{(s+1)} + \frac{K_2}{(s+2)}$$

$$K_1 = 2. \qquad K_2 = -2$$

$$f(t) = (2e^{-t} - 2e^{-2t})u(t)$$

Partial Fraction Expansion

Case 1. Roots of the Denominator of $F(s)$ Are Real and Distinct

Ex. given the following differential equations, find $f(t)$ if all initial values are zero

$$\frac{d^2 y}{dt^2} + 12 \frac{dy}{dt} + 32y = 32u(t)$$

$$s^2 Y(s) + 12sY(s) + 32Y(s) = \frac{32}{s}$$

Solving for the response, $Y(s)$, yields

$$Y(s) = \frac{32}{s(s^2 + 12s + 32)} = \frac{32}{s(s+4)(s+8)}$$

$$Y(s) = \frac{32}{s(s+4)(s+8)} = \frac{K_1}{s} + \frac{K_2}{(s+4)} + \frac{K_3}{(s+8)}$$

$$K_1 = \frac{32}{(s+4)(s+8)} \Big|_{s \rightarrow 0} = 1$$

$$K_2 = \frac{32}{s(s+8)} \Big|_{s \rightarrow -4} = -2$$

$$K_3 = \frac{32}{s(s+4)} \Big|_{s \rightarrow -8} = 1$$

$$Y(s) = \frac{1}{s} - \frac{2}{(s+4)} + \frac{1}{(s+8)}$$

$$y(t) = (1 - 2e^{-4t} + e^{-8t})u(t)$$

Partial Fraction Expansion

Case 2. Roots of the Denominator of $F(s)$ Are Real and Repeated

Given that the repeated roots are of multiplicity r at $-p_1$

$$\begin{aligned} F(s) &= \frac{N(s)}{D(s)} = \frac{N(s)}{(s+p_1)^r (s+p_2) \cdots (s+p_n)} \\ &= \frac{K_1}{(s+p_1)^r} + \frac{K_2}{(s+p_1)^{r-1}} + \cdots + \frac{K_r}{(s+p_1)} + \frac{K_{r+1}}{(s+p_2)} + \cdots + \frac{K_n}{(s+p_n)} \end{aligned}$$

To find K_1 through K_r for the roots of multiplicity greater than unity, we first multiply the equation by $(s+p_1)^r$.

$$\begin{aligned} F_1(s) &= (s+p_1)^r F(s) = \frac{(s+p_1)^r N(s)}{(s+p_1)^r (s+p_2) \cdots (s+p_n)} \\ &= K_1 + (s+p_1)K_2 + (s+p_1)^2 K_3 + \cdots + (s+p_1)^{r-1} K_r \\ &\quad + \frac{K_{r+1}(s+p_1)^r}{(s+p_2)} + \cdots + \frac{K_n(s+p_1)^r}{(s+p_n)} \end{aligned}$$

- We can solve for K_1 if we let s approach $-p_1$.
- We can solve for K_2 if we **differentiate** the eq. with respect to s and then let s approach $-p_1$.
- The general expression is:

$$K_i = \frac{1}{(i-1)!} \left. \frac{d^{i-1} F_1(s)}{ds^{i-1}} \right|_{s \rightarrow -p_1}$$
$$i = 1, 2, \dots, r; \quad 0! = 1$$

Partial Fraction Expansion

Case 2. Roots of the Denominator of $F(s)$ Are Real and Repeated

Ex. The roots of $(s + 2)^2$ in the denominator are repeated, since the factor is raised to an integer power higher than 1. In this case, the denominator root at 2 is a multiple root of multiplicity 2.

$$F(s) = \frac{2}{(s + 1)(s + 2)^2}$$

Partial fraction expansion is as:

$$F(s) = \frac{2}{(s + 1)(s + 2)^2} = \frac{K_1}{(s + 1)} + \frac{K_2}{(s + 2)^2} + \frac{K_3}{(s + 2)}$$

Note that each multiple root generates additional terms consisting of denominator factors of reduced multiplicity.

$K_1 = 2$ is found as previously described.

K_2 can be isolated by multiplying the equation by $(s + 2)^2$.

$$\frac{2}{s + 1} = (s + 2)^2 \frac{K_1}{(s + 1)} + K_2 + (s + 2)K_3$$

letting $s \rightarrow -2$; $K_2 = -2$.

To find K_3 we differentiate the above equation with respect to s ,

$$\frac{-2}{(s + 1)^2} = \frac{(s + 2)s}{(s + 1)^2} K_1 + K_3$$

K_3 is isolated and can be found if we let $s \rightarrow -2$. Hence, $K_3 = -2$.

$$f(t) = 2e^{-t} - 2te^{-2t} - 2e^{-2t}$$

Partial Fraction Expansion

Case 2. Roots of the Denominator of $F(s)$ Are Real and Repeated

Ex. For previous example, there is another way to find K_3 after finding K_1 and K_2 .

$$F(s) = \frac{2}{(s+1)(s+2)^2} = \frac{K_1}{(s+1)} + \frac{K_2}{(s+2)^2} + \frac{K_3}{(s+2)} , \quad K_1 = 2, K_2 = -2$$

Letting $s \rightarrow 0$ (any value rather than the roots), K_3 can be achieved: $K_3 = -2$

Thus,

$$f(t) = 2e^{-t} - 2te^{-2t} - 2e^{-2t}$$

Partial Fraction Expansion

Case 3. Roots of the Denominator of $F(s)$ Are Complex or Imaginary

- Given an $F(s)$ whose denominator has complex or purely imaginary roots

$$\begin{aligned} F(s) = \frac{N(s)}{D(s)} &= \frac{N(s)}{(s + p_1)(s^2 + as + b) \dots} \\ &= \frac{K_1}{(s + p_1)} + \frac{(K_2s + K_3)}{(s^2 + as + b)} + \dots \end{aligned}$$

where p_1 is real, and $(s^2 + as + b)$ has complex or purely imaginary roots.

- Here, K_2 and K_3 are found through balancing the coefficients of the equation after clearing fractions.

Partial Fraction Expansion

Case 3. Roots of the Denominator of $F(s)$ Are Complex or Imaginary

Using the items available in Laplace tables, we have

$$\mathcal{L}[Ae^{-at}\cos \omega t] = \frac{A(s+a)}{(s+a)^2 + \omega^2}$$

$$\mathcal{L}[Be^{-at}\sin \omega t] = \frac{B\omega}{(s+a)^2 + \omega^2}$$

Summing them gives

$$\mathcal{L}[Ae^{-at}\cos \omega t + Be^{-at}\sin \omega t] = \frac{A(s+a) + B\omega}{(s+a)^2 + \omega^2}$$

In this case of roots, we convert the denominator term related to the complex root to the form suggested by above equation by **completing the squares** in the denominator and adjusting terms in the numerator without changing its value.

Partial Fraction Expansion

Case 3. Roots of the Denominator of $F(s)$ Are Complex or Imaginary

Ex.

$$F(s) = \frac{3}{s(s^2 + 2s + 5)}$$

$$\frac{3}{s(s^2 + 2s + 5)} = \frac{K_1}{s} + \frac{K_2s + K_3}{s^2 + 2s + 5} \quad (2.31)$$

K_1 is found in the usual way to be $\frac{3}{5}$. K_2 and K_3 can be found by first multiplying Eq. (2.31) by the lowest common denominator, $s(s^2 + 2s + 5)$, and clearing the fractions. After simplification with $K_1 = \frac{3}{5}$, we obtain

$$3 = \left(K_2 + \frac{3}{5}\right)s^2 + \left(K_3 + \frac{6}{5}\right)s + 3$$

Balancing coefficients, $\left(K_2 + \frac{3}{5}\right) = 0$ and $\left(K_3 + \frac{6}{5}\right) = 0$. Hence $K_2 = -\frac{3}{5}$ and $K_3 = -\frac{6}{5}$. Thus,

$$F(s) = \frac{3}{s(s^2 + 2s + 5)} = \frac{3/5}{s} - \frac{3}{5} \frac{s+2}{s^2 + 2s + 5}$$

$$F(s) = \frac{3/5}{s} - \frac{3}{5} \frac{(s+1) + (1/2)(2)}{(s+1)^2 + 2^2}$$


$$f(t) = \frac{3}{5} - \frac{3}{5}e^{-t} \left(\cos 2t + \frac{1}{2} \sin 2t \right)$$

Partial Fraction Expansion

Case 3. Roots of the Denominator of $F(s)$ Are Complex or Imaginary

Alternative method is the same technique used for the $F(s)$ with real roots in the denominator. However, the residues of the complex and imaginary roots are themselves **complex conjugates**. Then, after taking the inverse Laplace transform, the resulting terms can be identified as

$$\frac{e^{j\theta} + e^{-j\theta}}{2} = \cos \theta$$

$$\frac{e^{j\theta} - e^{-j\theta}}{2j} = \sin \theta$$

Partial Fraction Expansion

Case 3. Roots of the Denominator of $F(s)$ Are Complex or Imaginary

Ex.
$$F(s) = \frac{3}{s(s^2 + 2s + 5)}$$
$$= \frac{3}{s(s + 1 + j2)(s + 1 - j2)}$$
$$= \frac{K_1}{s} + \frac{K_2}{s + 1 + j2} + \frac{K_3}{s + 1 - j2}$$

We achieved that $K_1 = 3/5$, Finding K_2 :

$$K_2 = \left. \frac{3}{s(s + 1 - j2)} \right|_{s \rightarrow -1-j2} = -\frac{3}{20}(2 + j1)$$

Similarly, K_3 is found to be the complex conjugate of K_2 . Thus:

$$F(s) = \frac{3/5}{s} - \frac{3}{20} \left(\frac{2 + j1}{s + 1 + j2} + \frac{2 - j1}{s + 1 - j2} \right)$$

From which

$$\begin{aligned} f(t) &= \frac{3}{5} - \frac{3}{20} [(2 + j1)e^{-(1+j2)t} + (2 - j1)e^{-(1-j2)t}] \\ &= \frac{3}{5} - \frac{3}{20} e^{-t} \left[4 \left(\frac{e^{j2t} + e^{-j2t}}{2} \right) + 2 \left(\frac{e^{j2t} + e^{-j2t}}{2j} \right) \right] \\ &= \frac{3}{5} - \frac{3}{5} e^{-t} \left(\cos 2t + \frac{1}{2} \sin 2t \right) \end{aligned}$$

Partial Fraction Expansion

Case 3. Roots of the Denominator of $F(s)$ Are Complex or Imaginary

Ex. Note. In order to visualize the solution, an alternate form of $f(t)$, obtained by trigonometric identities, is preferable.

$$f(t) = \frac{3}{5} - \frac{3}{5}e^{-t} \left(\cos 2t + \frac{1}{2} \sin 2t \right)$$

Using the amplitudes of the cos and sin terms, we factor out $\sqrt{1^2 + (1/2)^2}$ from the term in parentheses and obtain

$$f(t) = \frac{3}{5} - \frac{3}{5} \sqrt{1^2 + (1/2)^2} e^{-t} \left(\frac{1}{\sqrt{1^2 + (1/2)^2}} \cos 2t + \frac{1/2}{\sqrt{1^2 + (1/2)^2}} \sin 2t \right)$$

Letting $1/\sqrt{1^2 + (1/2)^2} = \cos \phi$ and $(1/2)/\sqrt{1^2 + (1/2)^2} = \sin \phi$,

$$f(t) = \frac{3}{5} - \frac{3}{5} \sqrt{1^2 + (1/2)^2} e^{-t} (\cos \phi \cos 2t + \sin \phi \sin 2t) = 0.6 - 0.671 e^{-t} \cos (2t - \phi)$$

where $\phi = \arctan 0.5 = 26.57^\circ$.

Partial Fraction Expansion

Case 3. Roots of the Denominator of $F(s)$ Are Complex or Imaginary

Ex. Note. Trick!

$$F(s) = \frac{3}{s(s^2 + 2s + 5)}$$

$$F(s) = \frac{3/5}{s} - \frac{3}{20} \left(\frac{2 + j1}{s + 1 + j2} + \frac{2 - j1}{s + 1 - j2} \right)$$

$$f(t) = 0.6 - 0.671e^{-t}\cos(2t - \phi)$$

Transfer Function

- We are now ready to formulate the system representation by establishing a viable definition for a function that algebraically relates a system's output to its input.
- This function will allow separation of the input, system and output into three separate and distinct parts, unlike the differential equation.
- The function will also allow us to algebraically combine mathematical representations of subsystems to yield a total system representation.

$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \cdots + a_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \cdots + b_0 r(t)$$

- Taking the Laplace Transform of the above differential equation,

$$a_n s^n C(s) + a_{n-1} s^{n-1} C(s) + \cdots + a_0 C(s) + \text{init. cond. terms of } c(t) = b_m s^m R(s) + b_{m-1} s^{m-1} R(s) + \cdots + b_0 R(s) + \text{init. cond. terms of } r(t)$$

Transfer Function

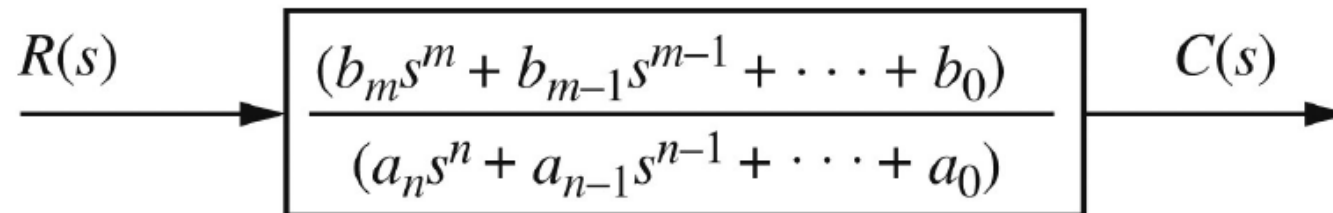
- For zero initial conditions,

$$\begin{aligned} & (a_n s^n + a_{n-1} s^{n-1} + \dots + a_0) C(s) \\ &= (b_m s^m + b_{m-1} s^{m-1} + \dots + b_0) R(s) \end{aligned}$$

- Constituting the ratio of out to input in the s -domain,

$$\frac{C(s)}{R(s)} = G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}$$

- Notice that this representation separates the output, $C(s)$, the input, $R(s)$, and the system.
- We call this ratio, $G(s)$, the transfer function and evaluate it with **zero initial conditions**.
- The transfer function can be represented as a block diagram, as shown below, with the input on the left, the output on the right, and the system transfer function inside the block.



Transfer Function

Ex. Find the transfer function represented by

$$\frac{dc(t)}{dt} + 2c(t) = r(t) \quad (2.55)$$

Taking the Laplace transform of both sides, assuming zero initial conditions, we have

$$sC(s) + 2C(s) = R(s) \quad (2.56)$$

$$G(s) = \frac{C(s)}{R(s)} = \frac{1}{s+2} \quad (2.57)$$

The output for step input, $C(s) = \frac{1}{s(s+2)}$

Use the result of Example 2.4 to find the response, $c(t)$ to an input, here a unit step, assuming zero initial conditions.

$$C(s) = \frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{1}{s+2} \quad \longrightarrow \quad c(t) = \frac{1}{2} (1 - e^{-2t})$$

Q2-1

Find the inverse Laplace transform of

$$F(s) = \frac{1}{s + 5}$$

a) $e^{5t}u(t)$

b) $te^{5t}u(t)$

c) $e^{-5t}u(t)$

d) $tu(t)$

e) None of the above

Q2-2

Ex.

PROBLEM: Find the inverse Laplace transform of $F(s) = 10/[s(s + 2)(s + 3)^2]$.

ANSWER:
$$f(t) = \frac{5}{9} - 5e^{-2t} + \frac{10}{3}te^{-3t} + \frac{40}{9}e^{-3t}$$